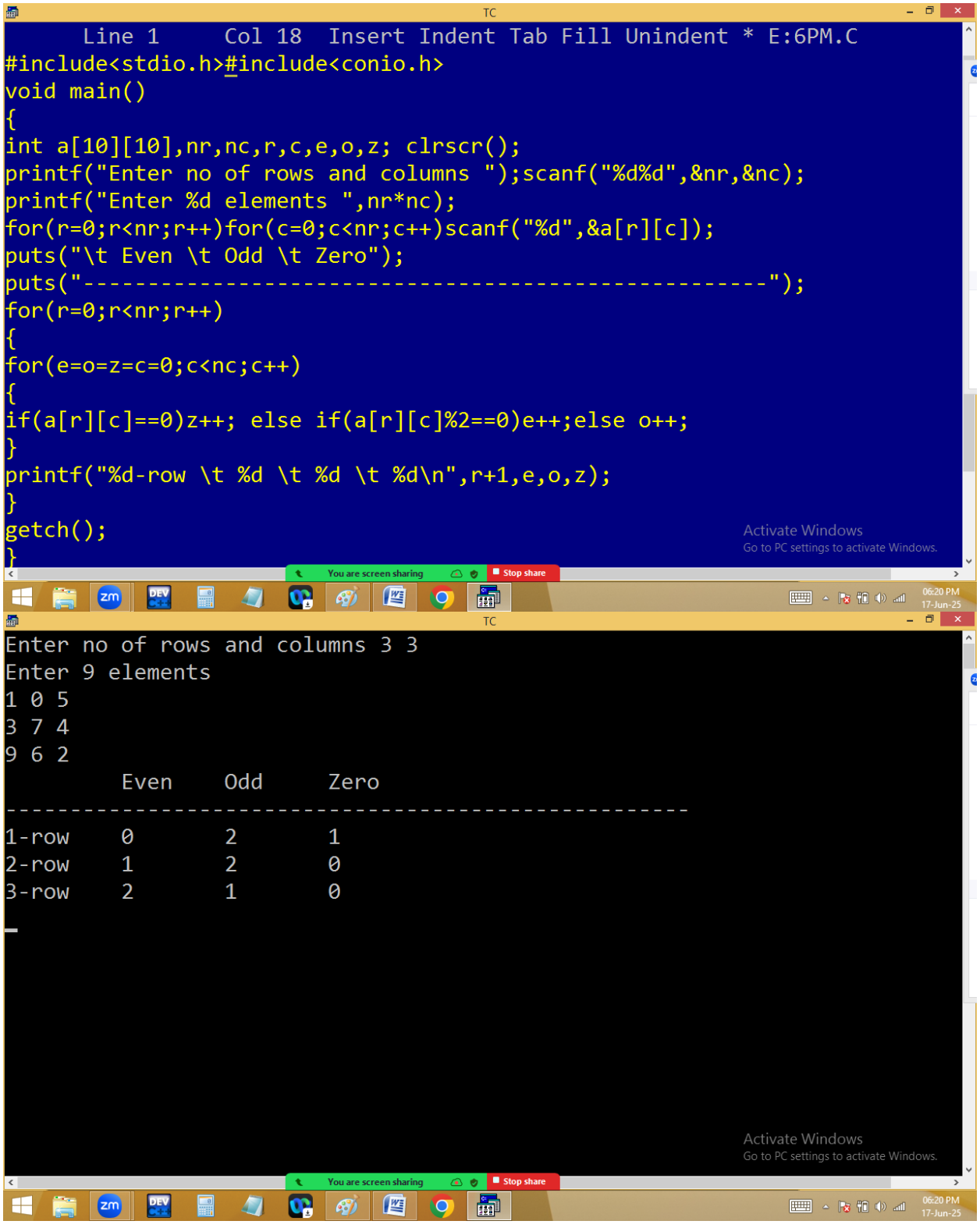
Finding no of even, odd, zero elements rowwise:



```
TC
      Line 1     Col 18  Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>#include<conio.h>
void main()
{
int a[10][10],nr,nc,r,c,e,o,z; clrscr();
printf("Enter no of rows and columns ");scanf("%d%d",&nr,&nc);
printf("Enter %d elements ",nr*nc);
for(r=0;r<nr;r++)for(c=0;c<nr;c++)scanf("%d",&a[r][c]);
puts("\t Even \t Odd \t Zero");
puts("-----------------------------------------------------");
for(r=0;r<nr;r++)
{
for(e=o=z=c=0;c<nc;c++)
{
if(a[r][c]==0)z++; else if(a[r][c]%2==0)e++;else o++;
}
printf("%d-row \t %d \t %d \t %d\n",r+1,e,o,z);
}
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

You are screen sharing Stop share

06:20 PM
17-Jun-25

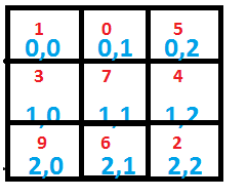
```
TC
Enter no of rows and columns 3 3
Enter 9 elements
1 0 5
3 7 4
9 6 2
         Even    Odd     Zero
-----------------------------------------------------
1-row    0       2       1
2-row    1       2       0
3-row    2       1       0
```

Activate Windows
Go to PC settings to activate Windows.

You are screen sharing Stop share

06:20 PM
17-Jun-25

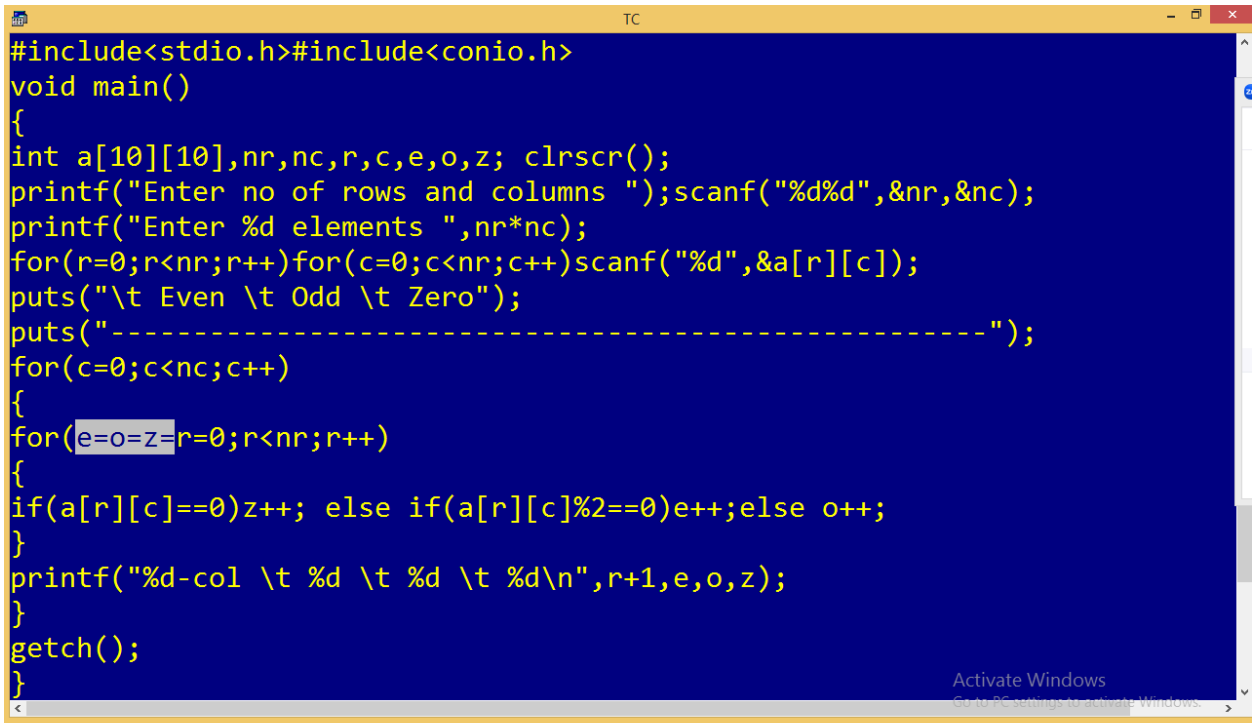
```
puts("\t Even \t Odd\tZero");
puts("-----------------------------------------------");
for( r=0; r<3;r++)
{
for(e=o=z=c=0; c<3;c++)
{
if(a[r][c]==0)z++;else if(a[r][c]%2==0)e++;else o++;
}
p("%d-row\t%d\t%d\t%d\n",r+1,e,o,z);
}
```

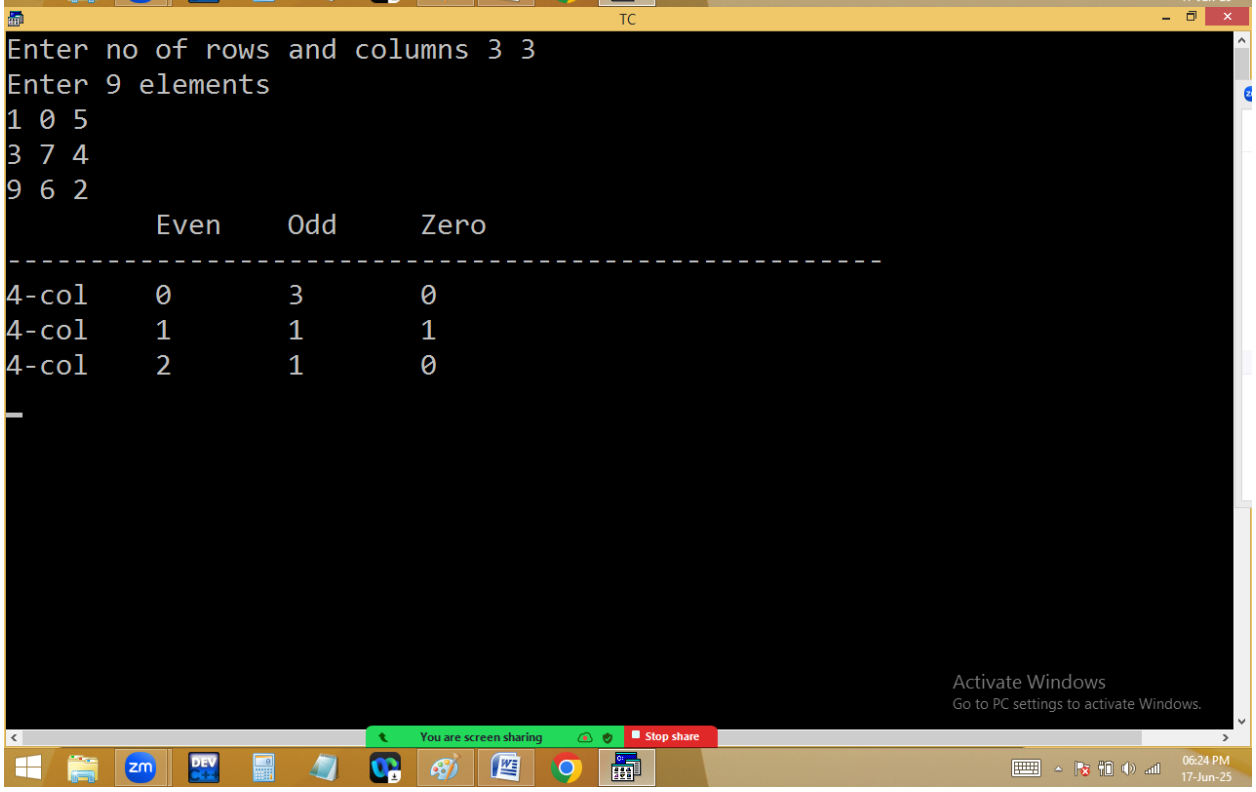




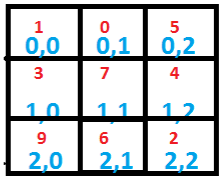
	Even	Odd	Zero
1-row	0	2	1
2-row	1	2	0
3-row	2	1	0

Column wise:


TC
#include<stdio.h>#include<conio.h>
void main()
{
int a[10][10],nr,nc,r,c,e,o,z; clrscr();
printf("Enter no of rows and columns ");scanf("%d%d",&nr,&nc);
printf("Enter %d elements ",nr*nc);
for(r=0;r<nr;r++)for(c=0;c<nr;c++)scanf("%d",&a[r][c]);
puts("\t Even \t Odd \t Zero");
puts("---");
for(c=0;c<nc;c++)
{
for(e=o=z=r=0;r<nr;r++)
{
if(a[r][c]==0)z++; else if(a[r][c]%2==0)e++;else o++;
}
printf("%d-col \t %d \t %d \t %d\n",r+1,e,o,z);
}
getch();
}
Activate Windows
You are screen sharing
Stop share


TC
Enter no of rows and columns 3 3
Enter 9 elements
1 0 5
3 7 4
9 6 2
Even Odd Zero
4-col 0 3 0
4-col 1 1 1
4-col 2 1 0
Activate Windows
Go to PC settings to activate Windows.
You are screen sharing
Stop share

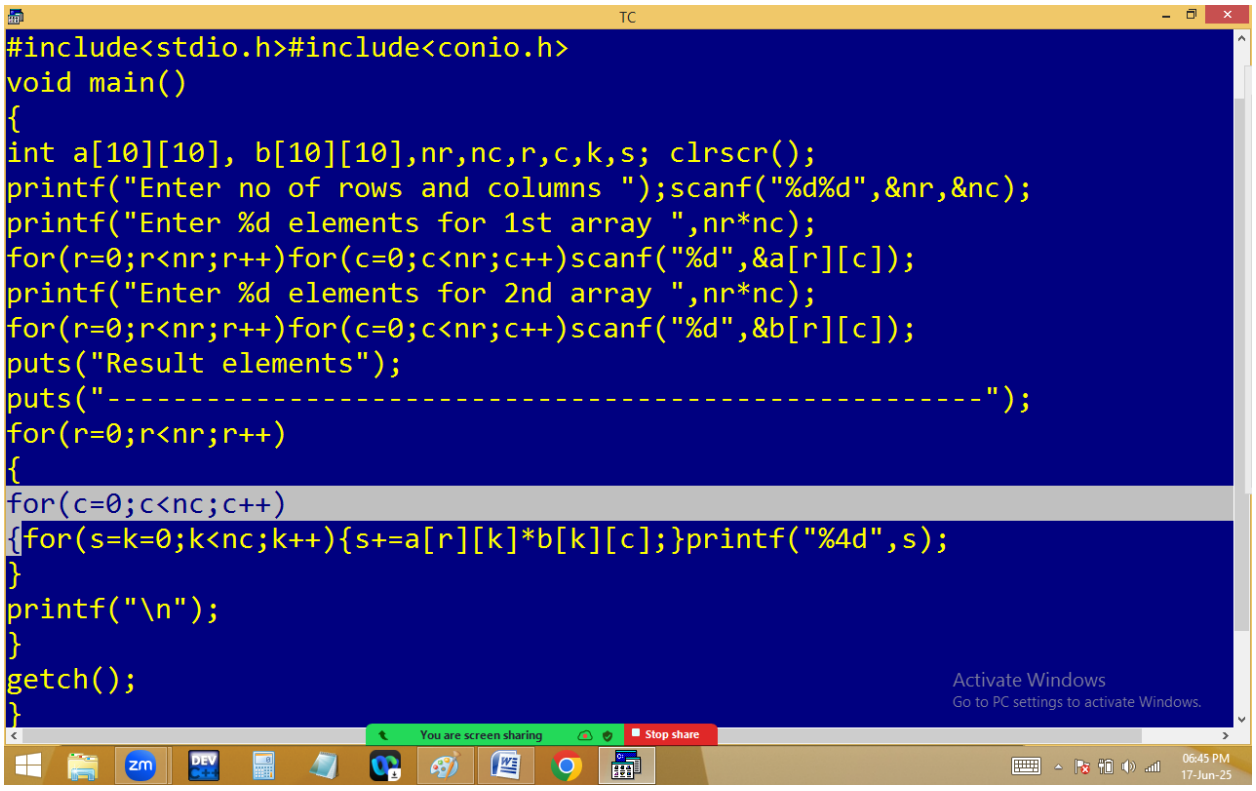
```
puts("\t Even \t Odd\tZero");
puts("-----------------------------------------------");
for(e=o=z=c=0; c<3;c++)
{
for( r=0; r<3;r++)
{  e=o=z=   0;
if(a[r][c]==0)z++;else if(a[r][c]%2==0)e++;else o++;
}
p("%d-col\t%d\t%d\t%d\n",r+1,e,o,z);
}
```

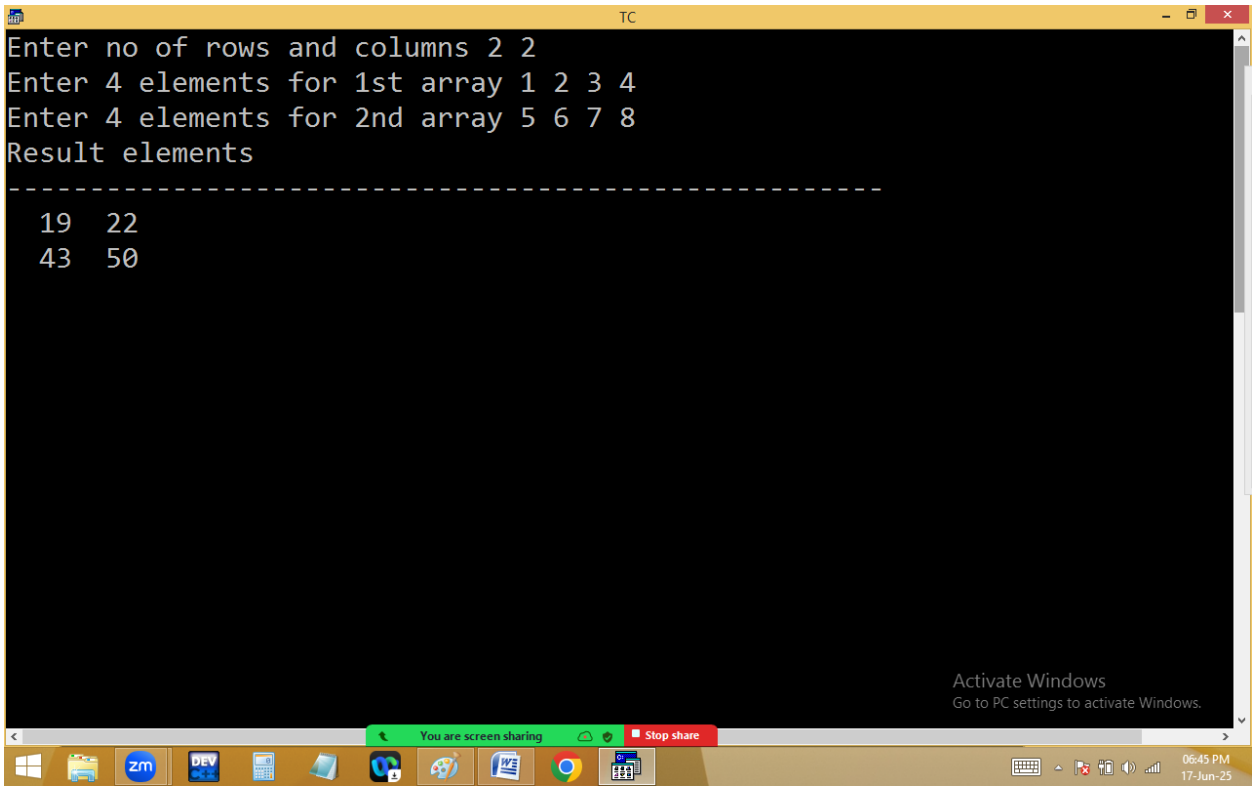


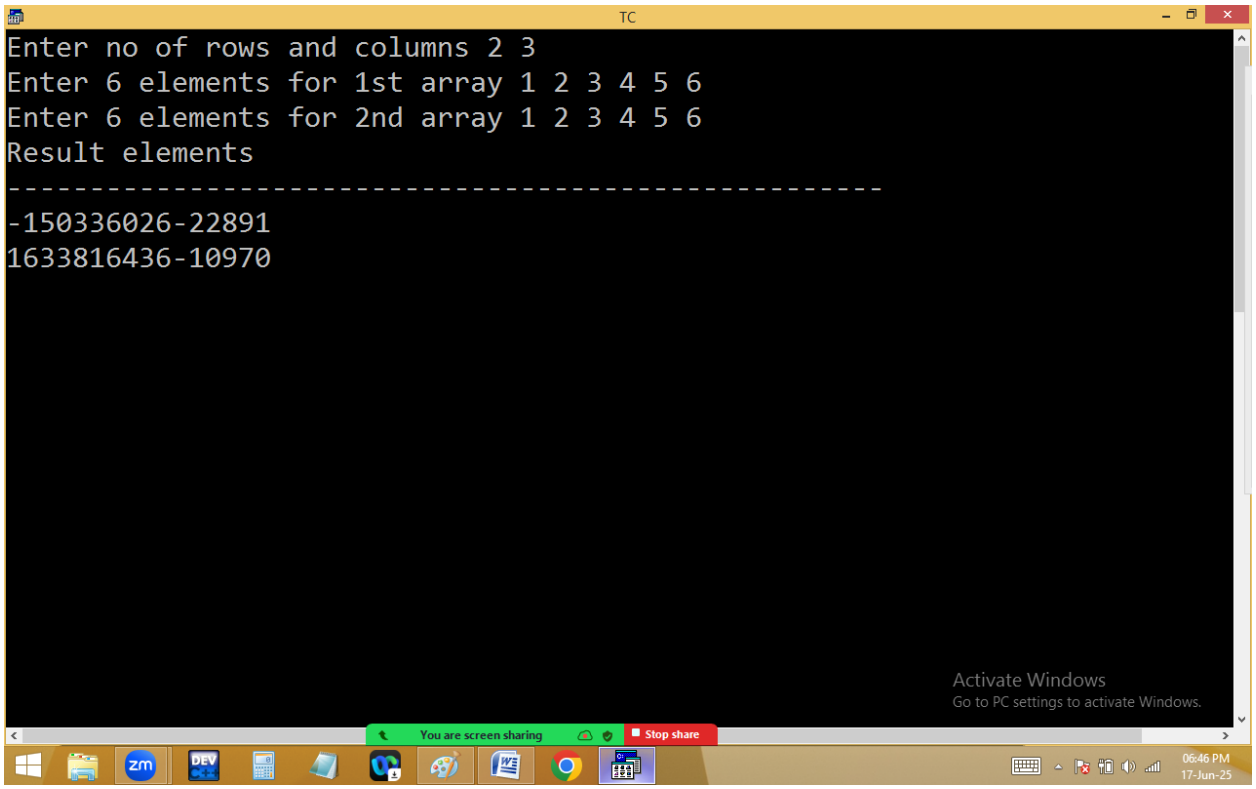


	Even	Odd	Zero
1-col	0	3	0
2-col	1	1	1
3-col	2	1	0

Matrix multiplication:

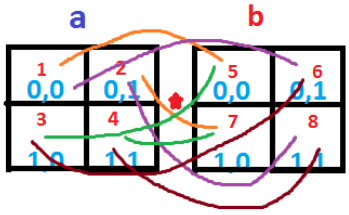

TC
#include<stdio.h>#include<conio.h>
void main()
{
int a[10][10], b[10][10],nr,nc,r,c,k,s; clrscr();
printf("Enter no of rows and columns ");scanf("%d%d",&nr,&nc);
printf("Enter %d elements for 1st array ",nr*nc);
for(r=0;r<nr;r++)for(c=0;c<nr;c++)scanf("%d",&a[r][c]);
printf("Enter %d elements for 2nd array ",nr*nc);
for(r=0;r<nr;r++)for(c=0;c<nr;c++)scanf("%d",&b[r][c]);
puts("Result elements");
puts("---");
for(r=0;r<nr;r++)
{
for(c=0;c<nc;c++)
{for(s=k=0;k<nc;k++){s+=a[r][k]*b[k][c];}printf("%4d",s);
}
printf("\n");
}
getch();
}
Activate Windows
Go to PC settings to activate Windows.
You are screen sharing
Stop share
06:45 PM
17-Jun-25


TC
Enter no of rows and columns 2 2
Enter 4 elements for 1st array 1 2 3 4
Enter 4 elements for 2nd array 5 6 7 8
Result elements
----
19 22
43 50
Activate Windows
Go to PC settings to activate Windows.
You are screen sharing
Stop share
06:45 PM
17-Jun-25



Note: 1st array cols and 2nd array rows should be same.

```
for(r=0;r<2;r++)
{
for(c=0;c<2;c++)
{  s=0;
for(k=0;k<2;k++)
{
s+=a[r][k]*b[k][c];
}
p( s );
}
p("\n");
}
```



r	c	k	s
0	0	012	0+1*5=5+ 2*7=19
0	1	012	0+1*6=6+2*8=22
1	0	012	0+3*5=15+4*7=43
1	1	012	0+3*6=18+4*8=50

3-dimensional arrays:

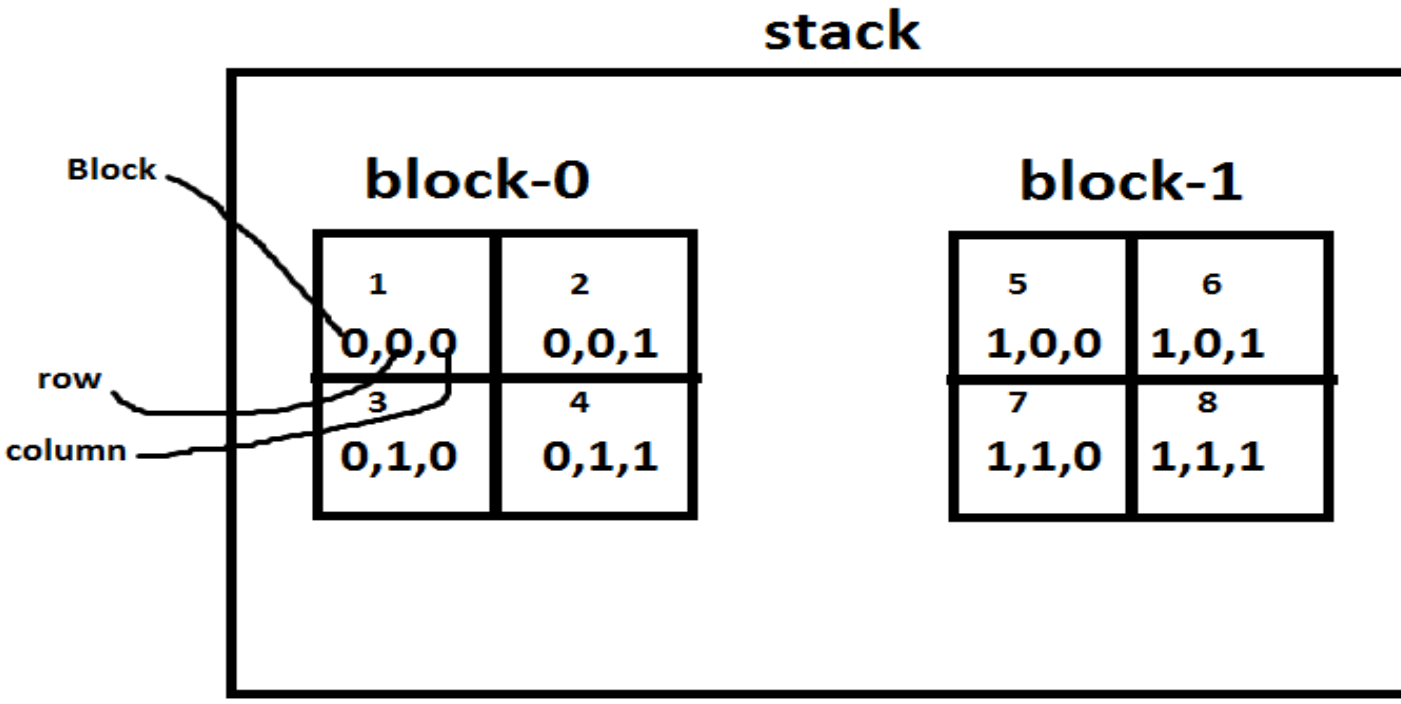
An array with several blocks, rows and columns.

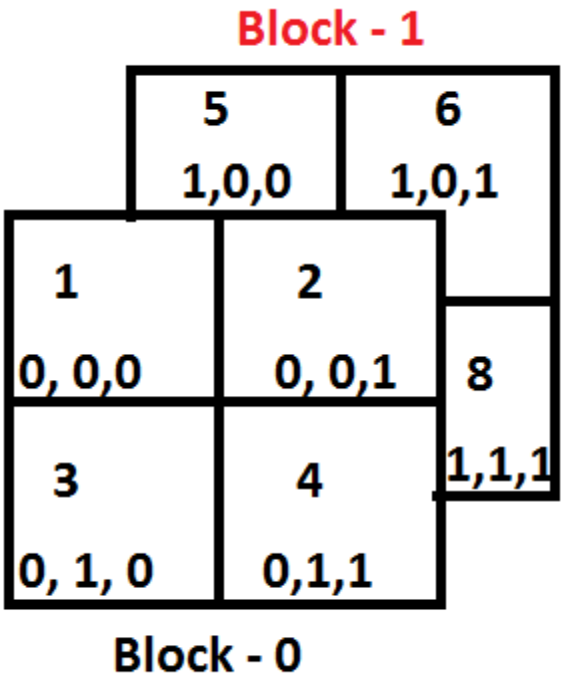
An array with 3 subscripting operators [ ] [ ] [ ].

Syntax:

datatype variable [blocks] [rows] [columns];

Eg: int a[2][2][2]={1,2,3,4,5,6,7,8};



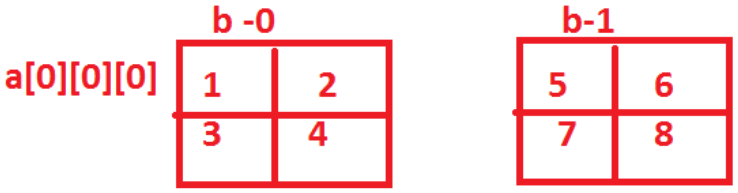


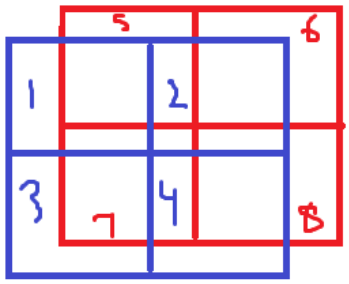
Eg:

rows

int a[2][2][2]; an array with 3 subscripts [ ].

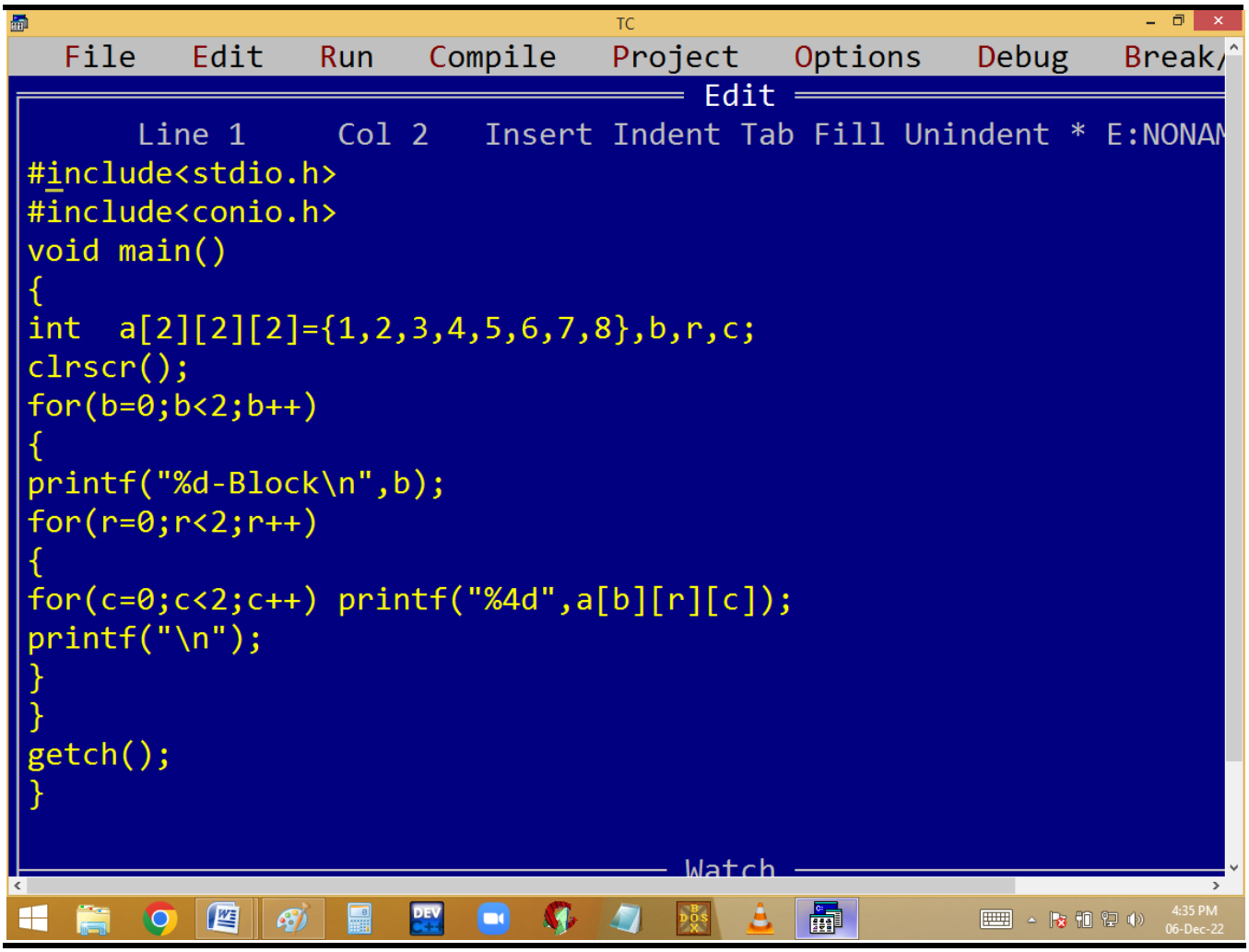
blocks columns

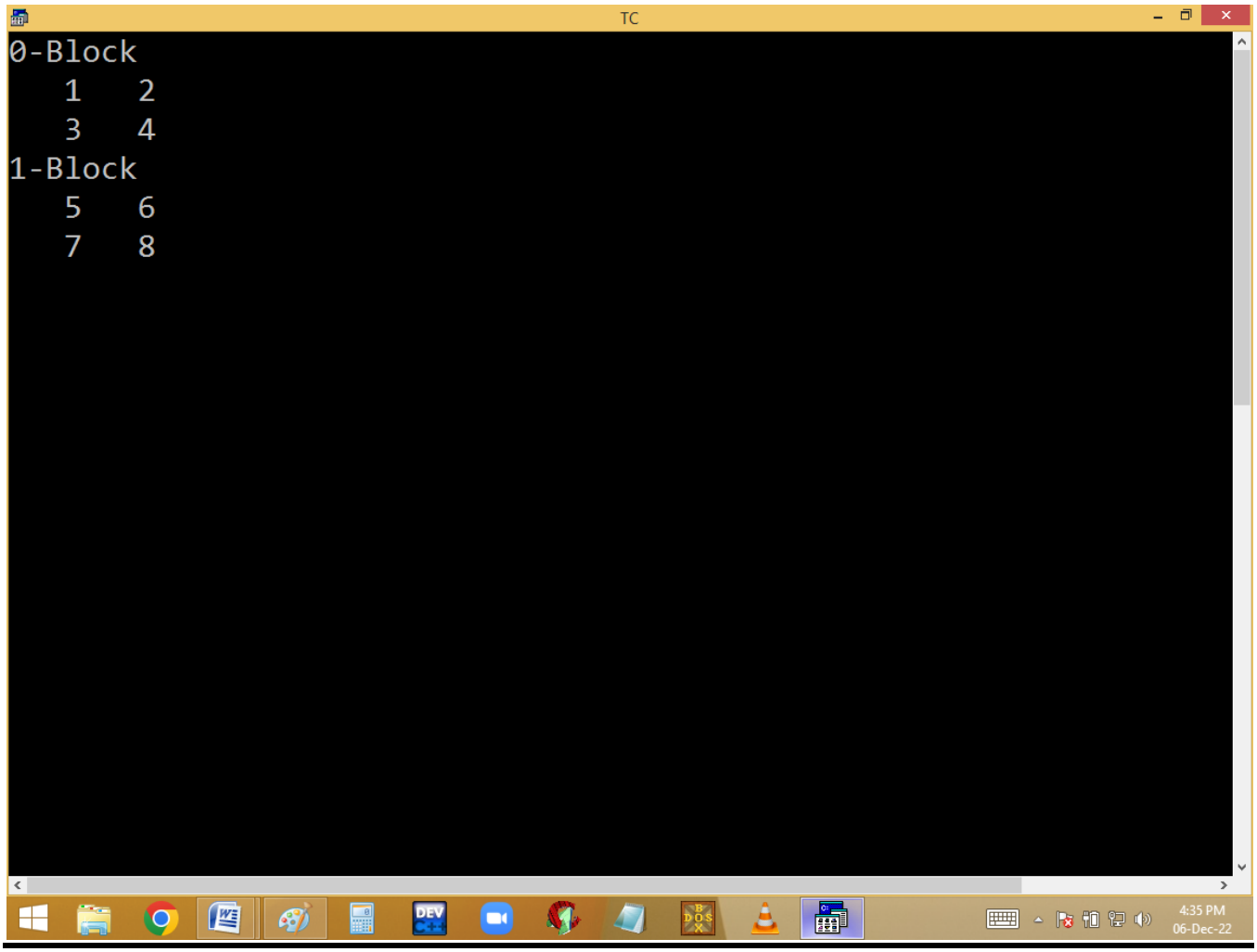




eg: int class[2][60][6];

datatype class[sections][stus][marks];





4-dimensional array:

An array with several sets, blocks, rows and columns.

An array with 4 subscripting operators [ ] [ ] [ ] [ ]

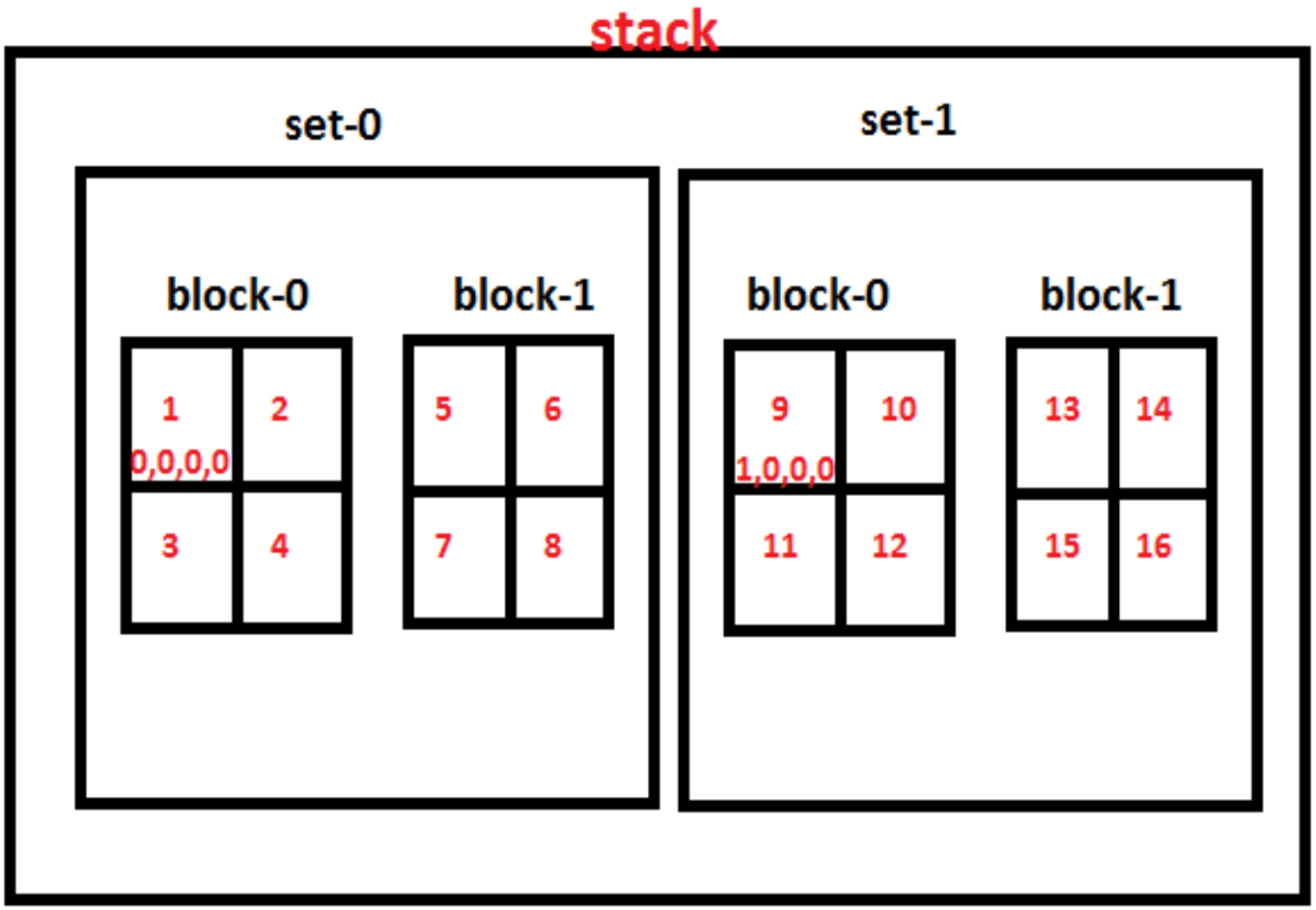
.

Syntax:

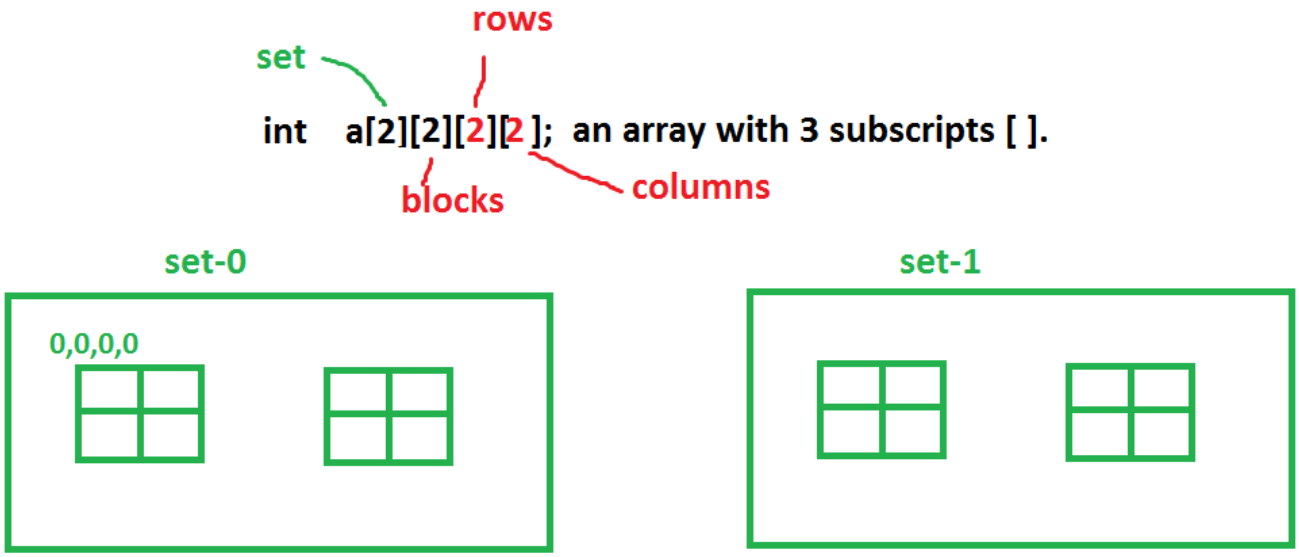
datatype variable [sets] [blocks] [rows] [cols];

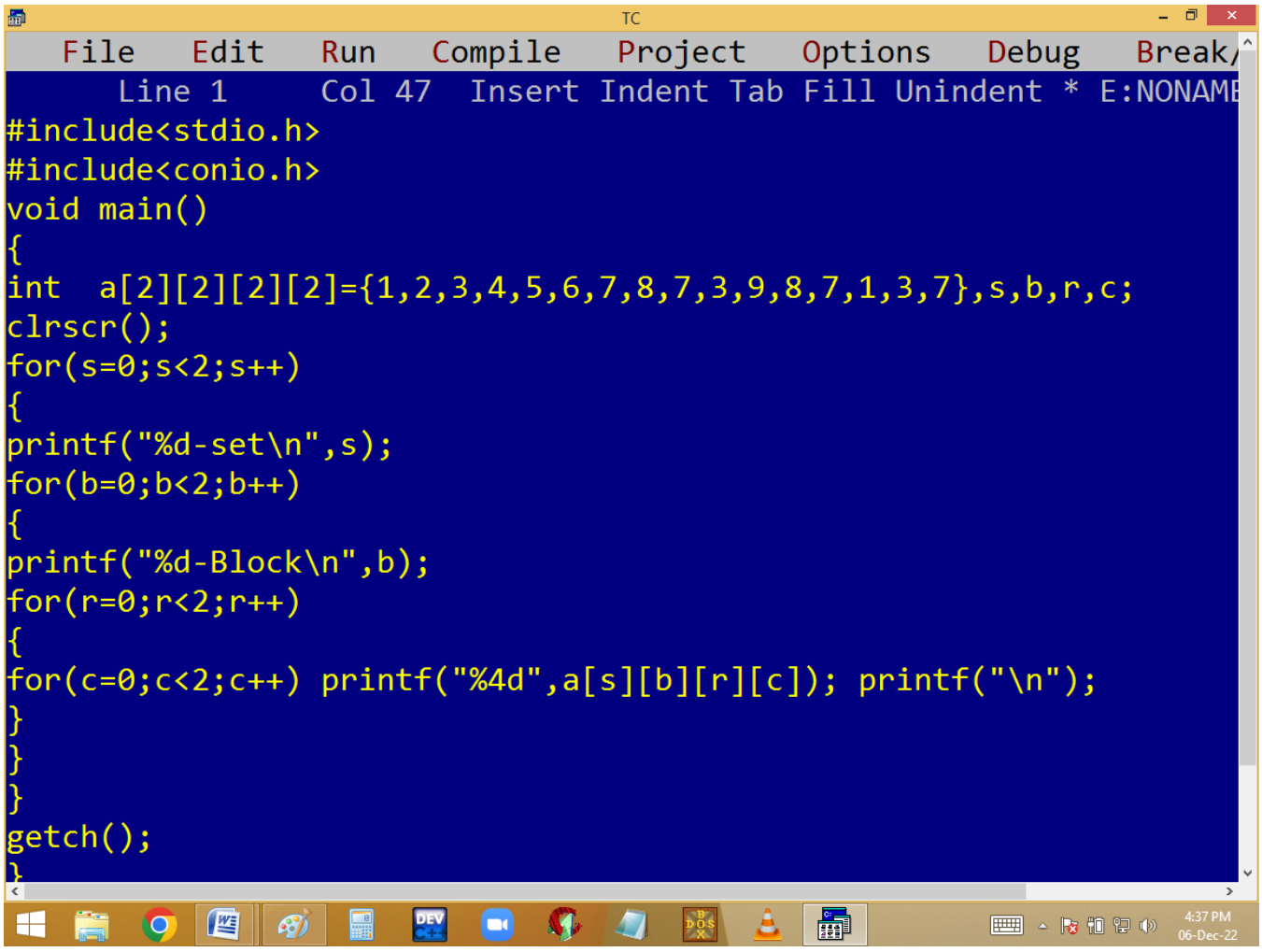
eg:

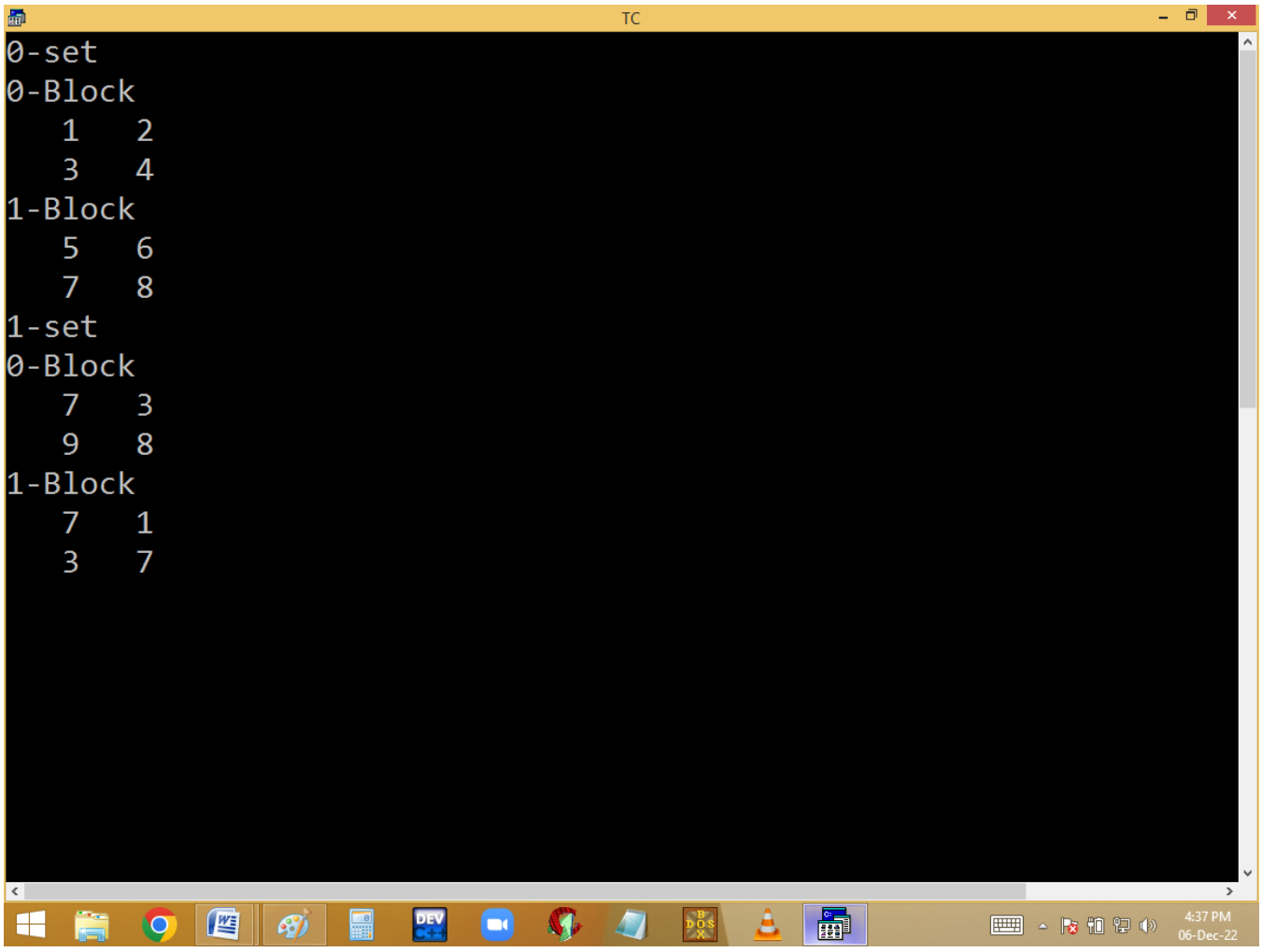
```
int a[2] [2] [2] [2]= {1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16};
```



datatype var[class][sec][stu][marks];
int school[5][2][60][6];




TC
File Edit Run Compile Project Options Debug Break/
Line 1 Col 47 Insert Indent Tab Fill Unindent * E:NONAME
#include<stdio.h>
#include<conio.h>
void main()
{
int a[2][2][2][2]={1,2,3,4,5,6,7,8,7,3,9,8,7,1,3,7},s,b,r,c;
clrscr();
for(s=0;s<2;s++)
{
printf("%d-set\n",s);
for(b=0;b<2;b++)
{
printf("%d-Block\n",b);
for(r=0;r<2;r++)
{
for(c=0;c<2;c++) printf("%4d",a[s][b][r][c]); printf("\n");
}
}
}
getch();
}
4:37 PM
06-Dec-22


TC
0-set
0-Block
1 2
3 4
1-Block
5 6
7 8
1-set
0-Block
7 3
9 8
1-Block
7 1
3 7
4:37 PM
06-Dec-22

STRINGS

- A group of characters is called string.
- It is one dimensional character array.
- It is alpha-numeric.
- It is an implicit pointer.
- It is a derived data type.

Note:

- **One byte should be left for Null char(\0). Otherwise we are getting garbage or junk values. Null char indicates string is completed.**

- **String variable Size can't be less than string. Otherwise we are getting error.**
- **Using = operator, we can't copy a string into another. We have to use strcpy() or copy character by character manually.**
- **Using == (comparison) operator, we can't compare two strings. Use strcmp() or compare the characters one by one manually.**

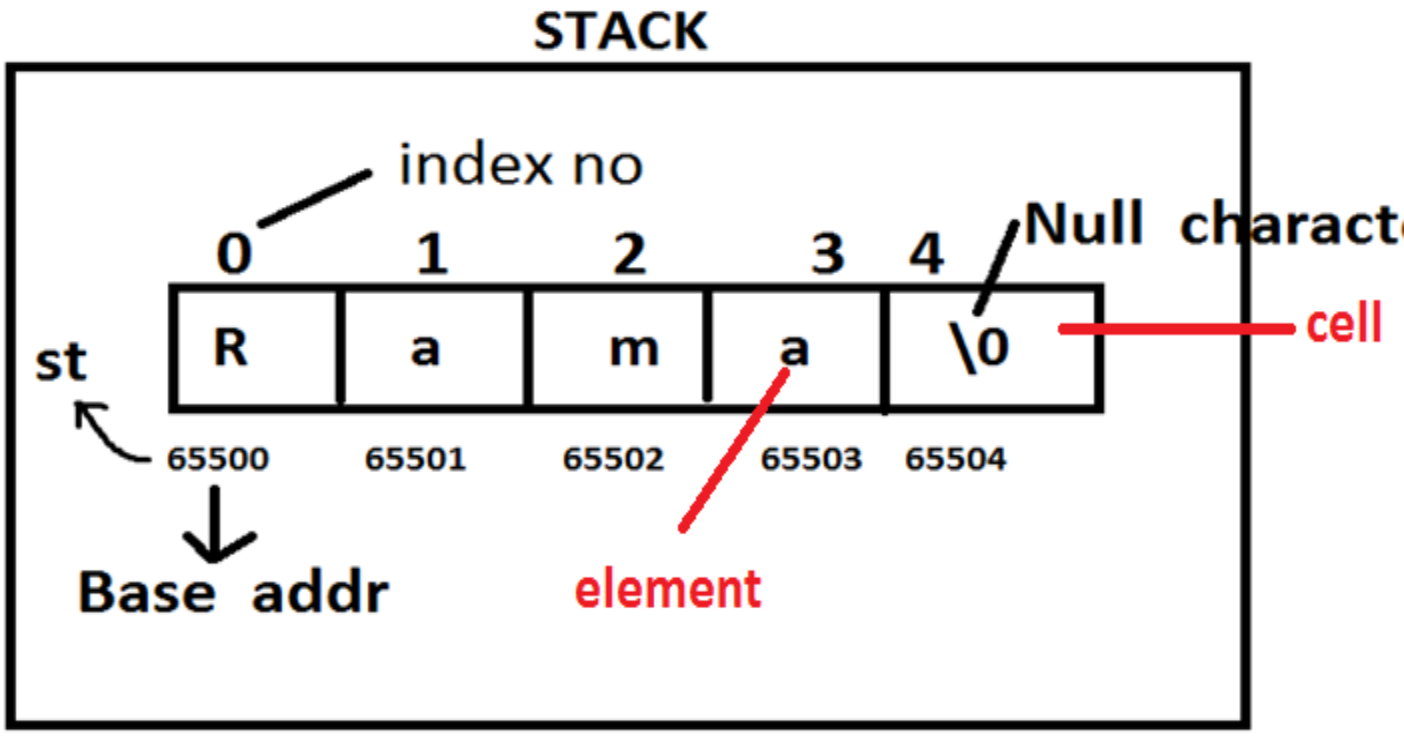
Syntax:

char variable [size] = "string";

or

char variable[ ]="string";

Eg: char st[5] = "Rama";



Note: String is implicit pointer because of string variable stores base address.

String declaration methods:

char st [5] =”rama”; Ok

char st [20] =” Naresh It”; Ok

char st [4] = { ‘r’, ‘a’, ‘m’ }; Ok → char array.

char st[3]= “ram”; It gives garbage values in printing.

char st [3] = “rama”; error

char st[0]; error

char st[0]=”abc”; Ok

char st[-5]; error

char st[5.5]; error

char st[5%3]; Ok ➔ char st[2];

char st[3+2]; ➔ st[5] ➔ Ok

char st[ ] =”Ram”; Ok.

char st[ ] ; error

int n=20;

char st[n]; No

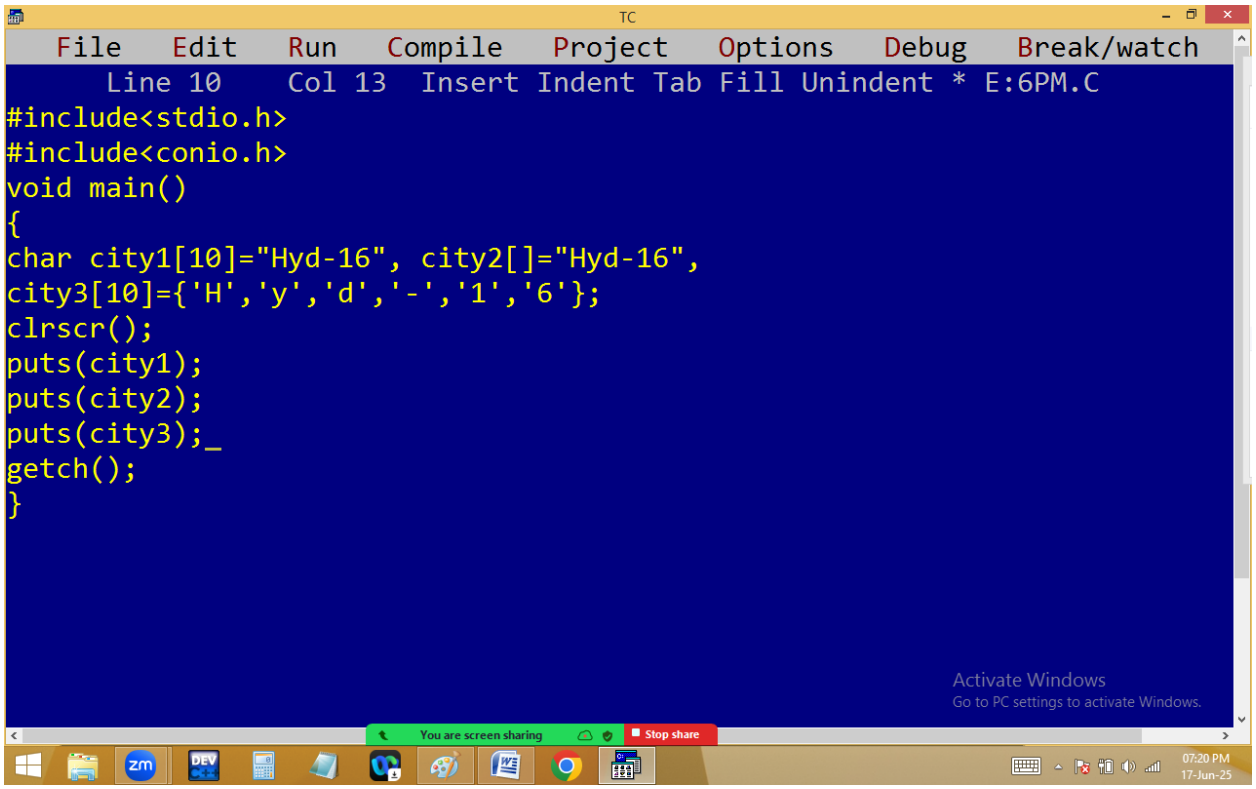
#define n 20

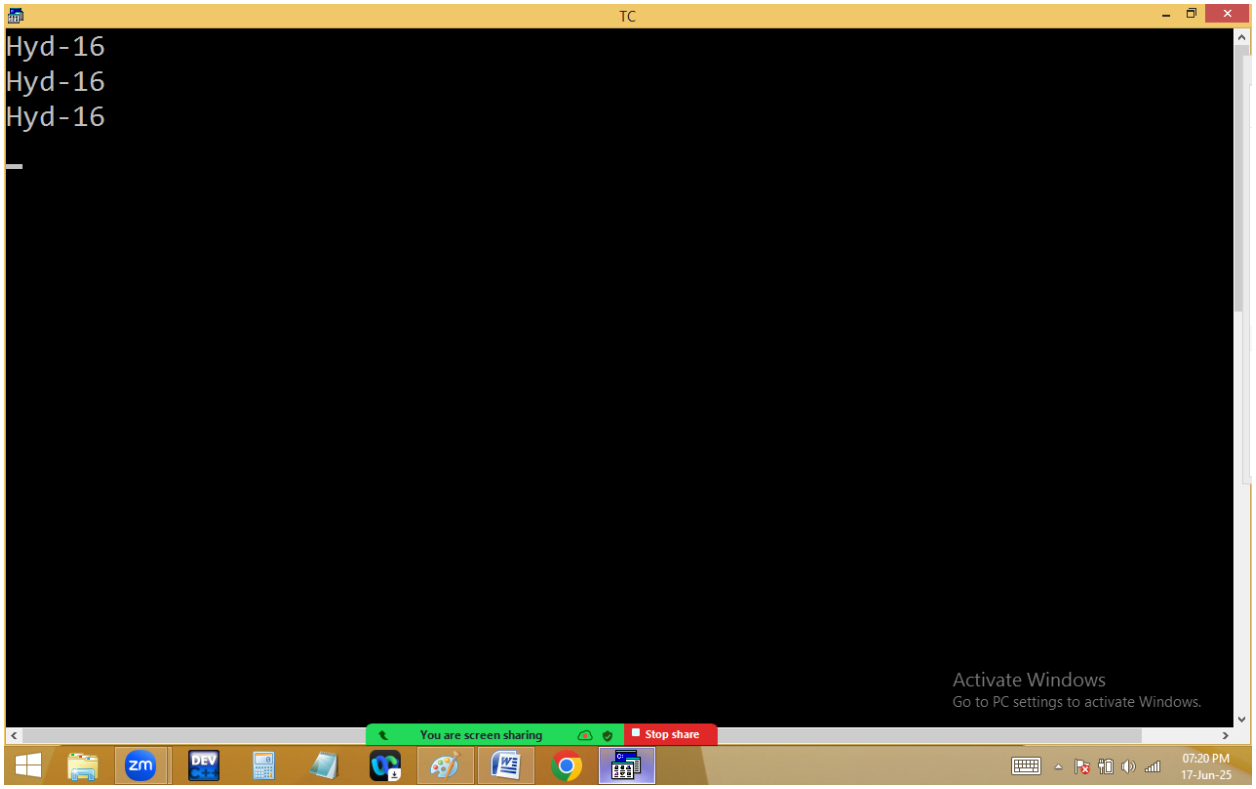
char st[n]; Ok

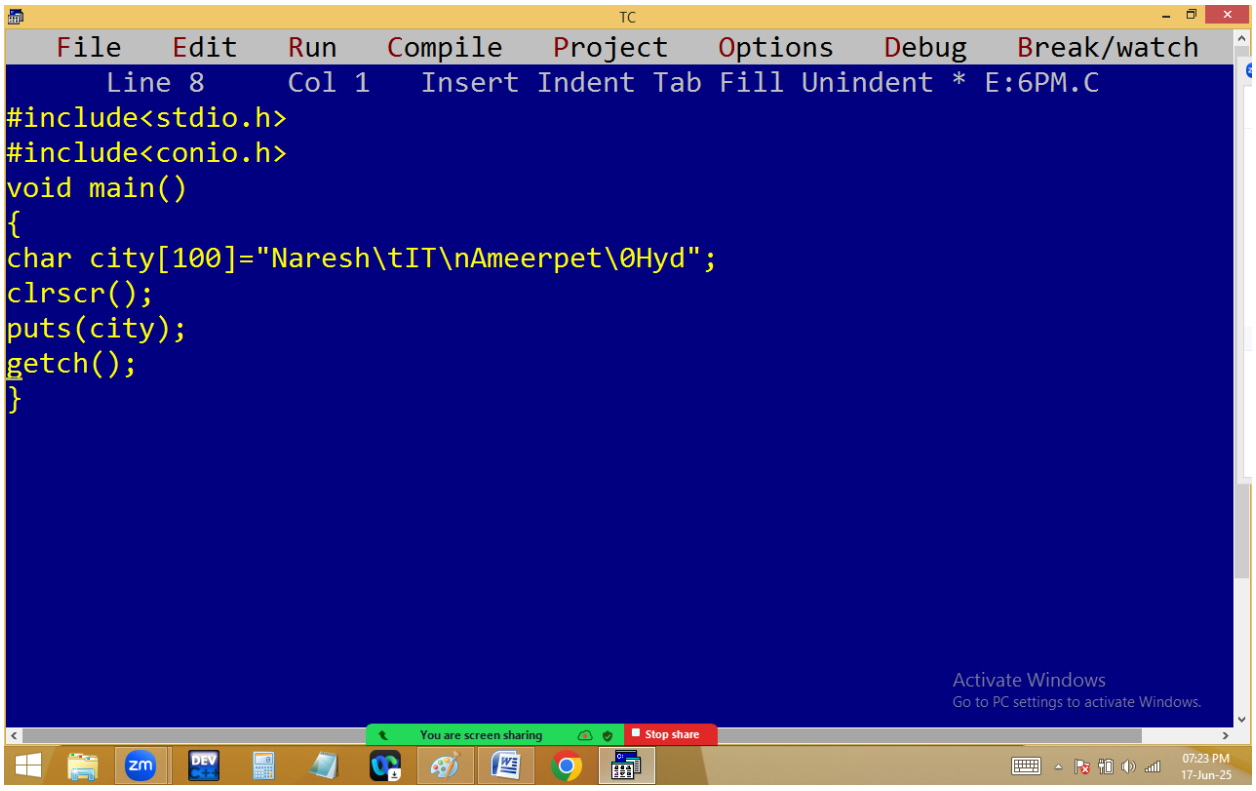
Note: String variable size always constant positive integer value.

Eg:

Direct initialization of a string:

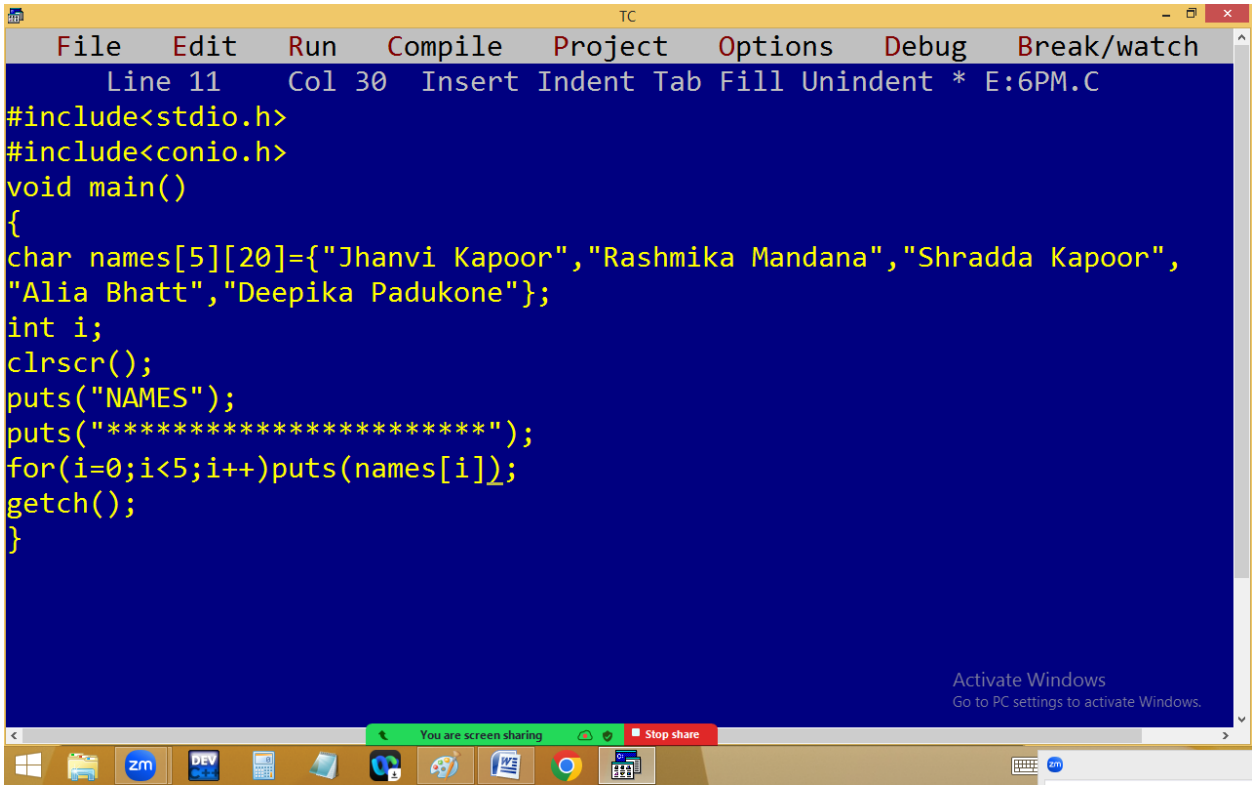

TC
File Edit Run Compile Project Options Debug Break/watch
Line 10 Col 13 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char city1[10]="Hyd-16", city2[]="Hyd-16",
city3[10]={'H','y','d','-','1','6'};
clrscr();
puts(city1);
puts(city2);
puts(city3);
getch();
}
Activate Windows
Go to PC settings to activate Windows.
You are screen sharing
Stop share
07:20 PM
17-Jun-25

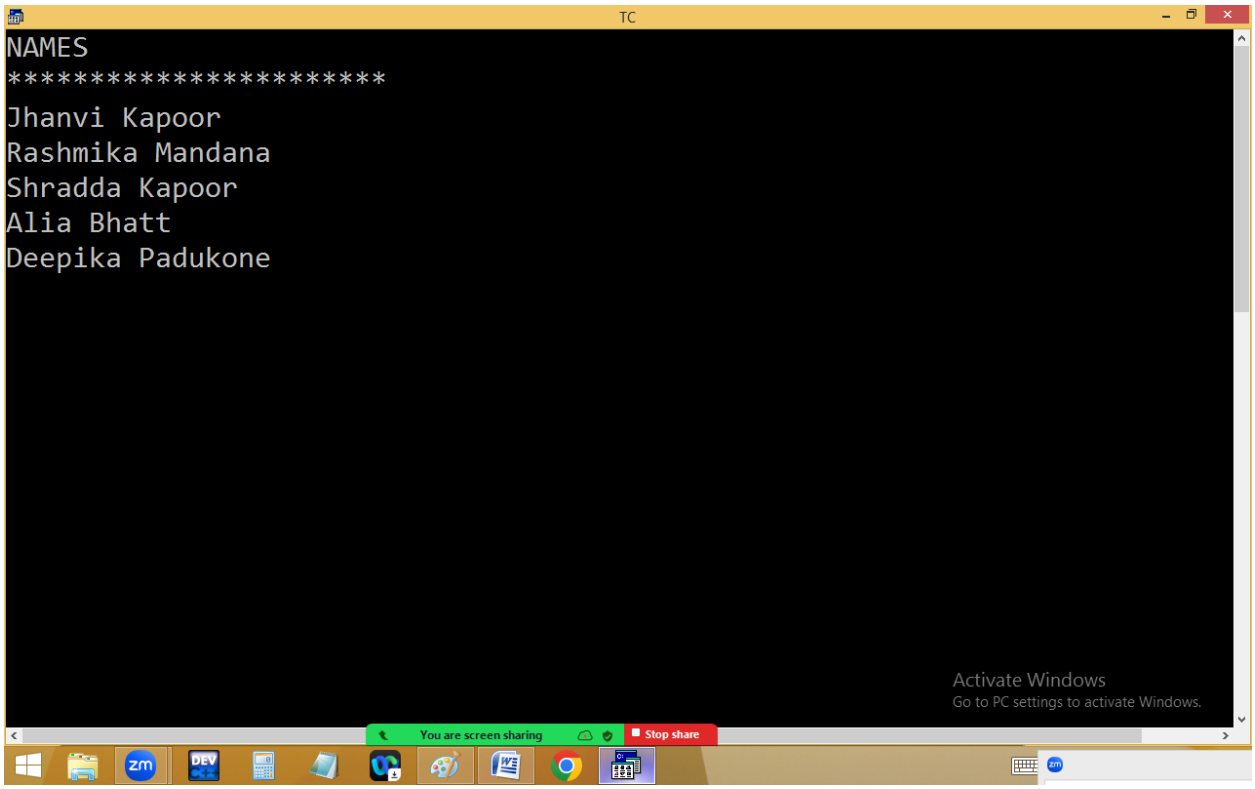

TC
Hyd-16
Hyd-16
Hyd-16
Activate Windows
Go to PC settings to activate Windows.
You are screen sharing
Stop share
07:20 PM
17-Jun-25


TC
File Edit Run Compile Project Options Debug Break/watch
Line 8 Col 1 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char city[100]="Naresh\tIT\nAmeerpet\0Hyd";
clrscr();
puts(city);
getch();
}
Activate Windows
Go to PC settings to activate Windows.
You are screen sharing
Stop share
07:23 PM
17-Jun-25

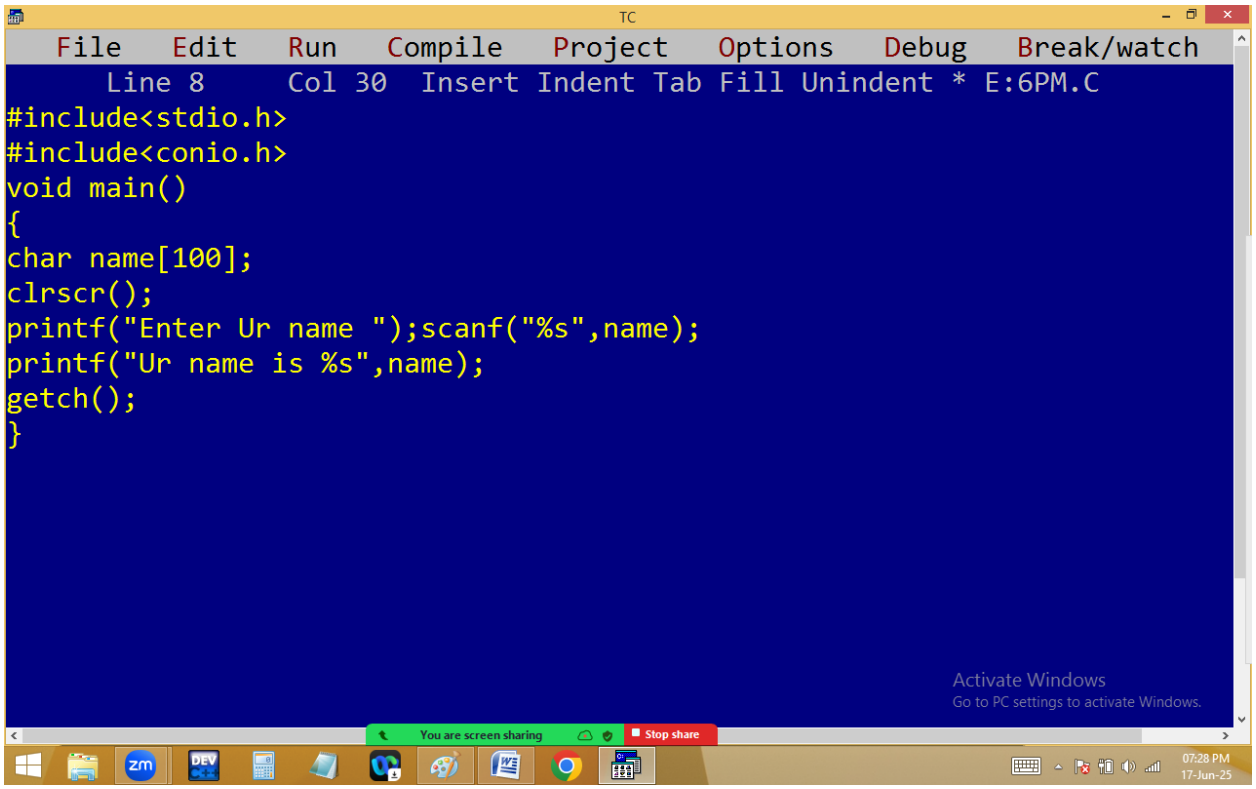

TC
Naresh IT
Ameerpet
You are screen sharing
Stop share
Activate Windows
Go to PC settings to activate Windows.
Mute
Start Video
Security
Participants
18
Chat
New share
Pause Share
Annotate
Slide Control
Apps
More
07:23 PM
17-Jun-25

Working with multiple strings:

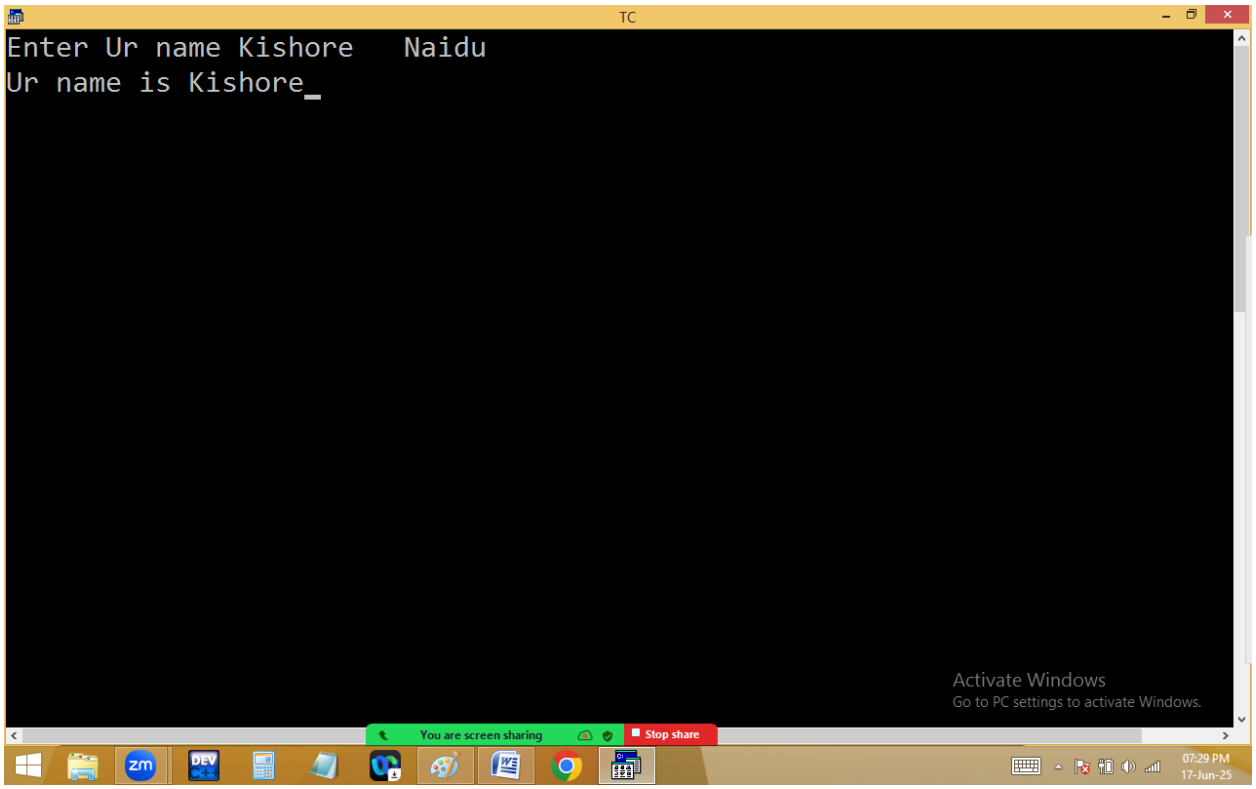




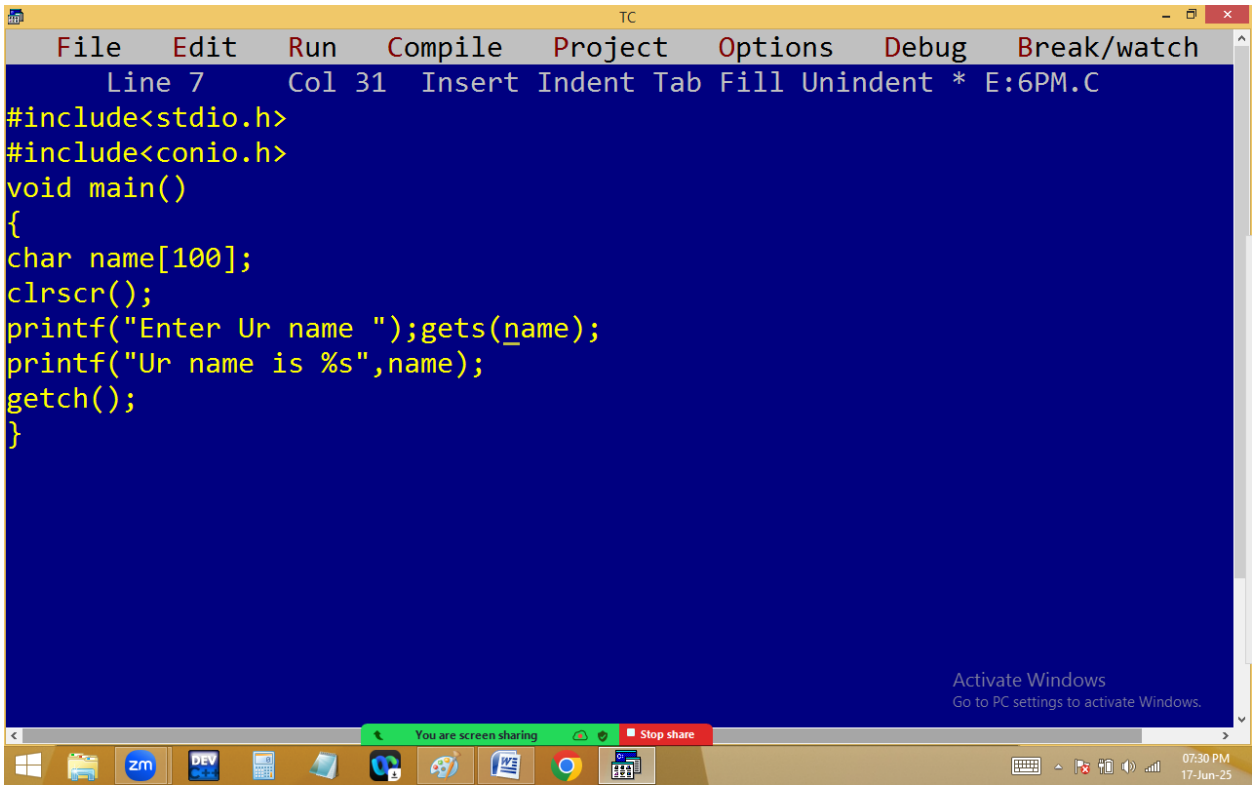
Reading and printing a string?

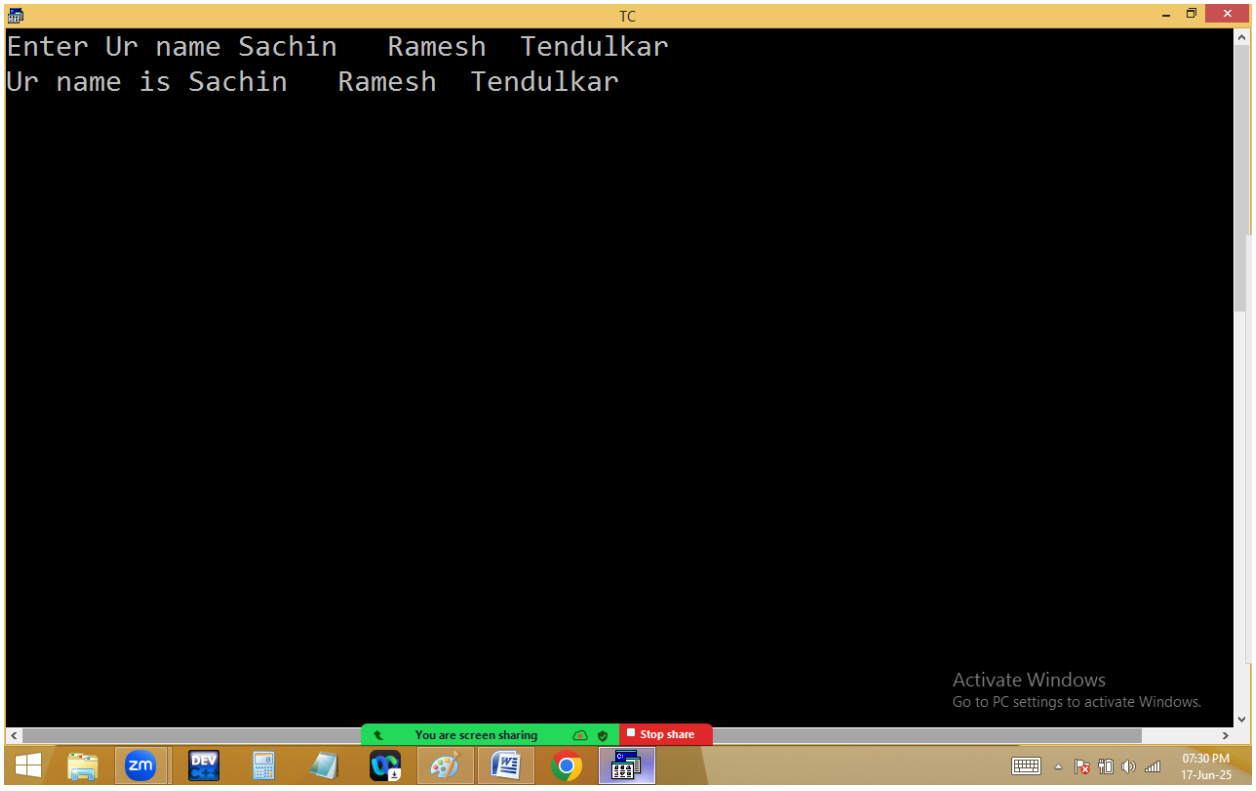

TC
File Edit Run Compile Project Options Debug Break/watch
Line 8 Col 30 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char name[100];
clrscr();
printf("Enter Ur name ");scanf("%s",name);
printf("Ur name is %s",name);
getch();
}
Activate Windows
Go to PC settings to activate Windows.
You are screen sharing
Stop share
07:28 PM
17-Jun-25

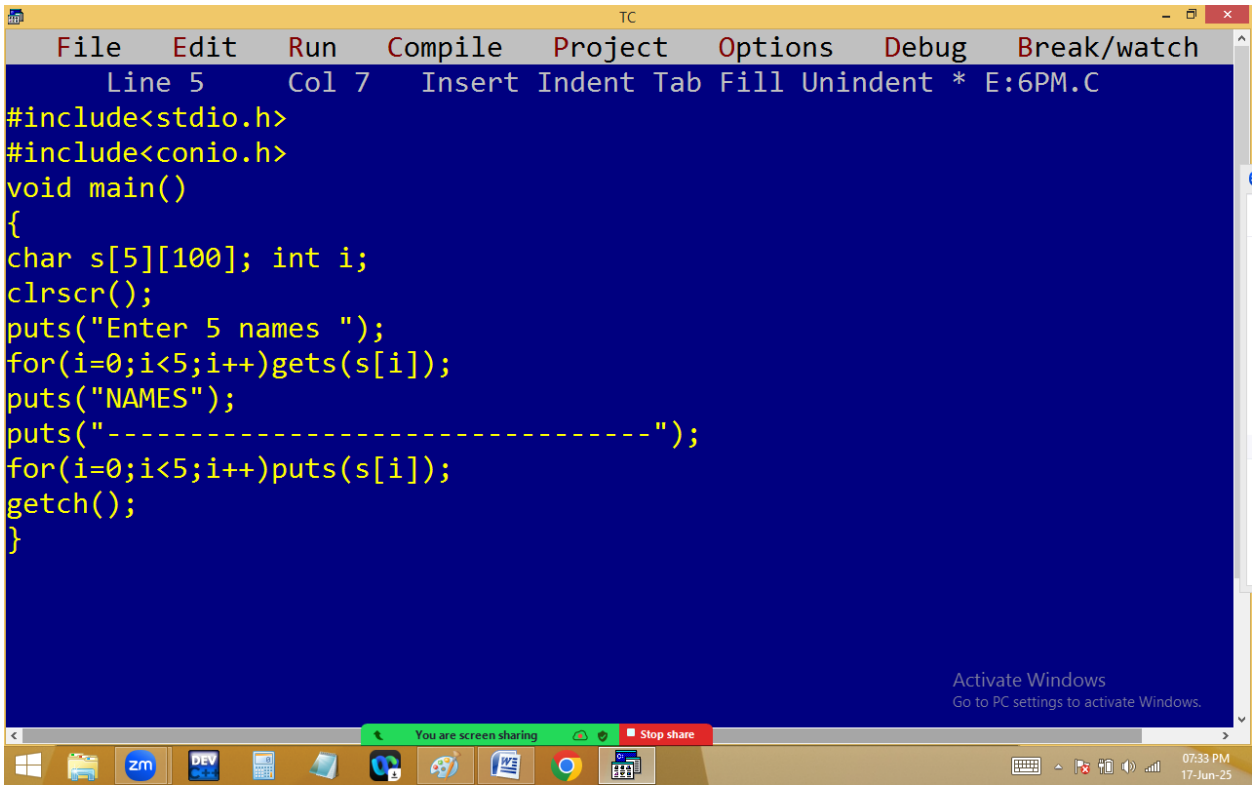

TC
Enter Ur name Kishore
Ur name is Kishore
You are screen sharing
Stop share
Activate Windows
Go to PC settings to activate Windows.
Mute
Start Video
Security
Participants
19
Chat
New share
Pause Share
Annotate
Slide Control
Apps
More
07:29 PM
17-Jun-25

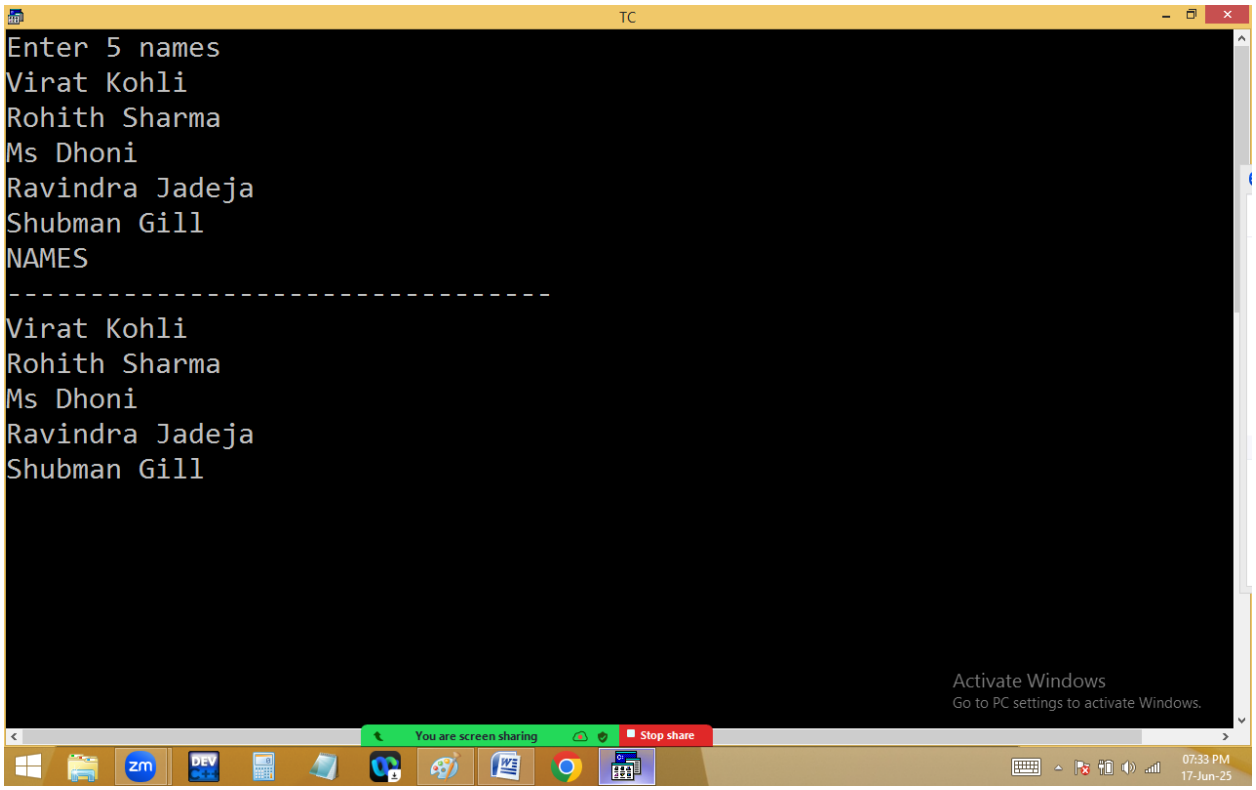


gets(): It read the string with spaces until pressing enter key.


TC
File Edit Run Compile Project Options Debug Break/watch
Line 7 Col 31 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char name[100];
clrscr();
printf("Enter Ur name ");gets(name);
printf("Ur name is %s",name);
getch();
}
Activate Windows
Go to PC settings to activate Windows.
You are screen sharing
Stop share
07:30 PM
17-Jun-25


TC
Enter Ur name Sachin Ramesh Tendulkar
Ur name is Sachin Ramesh Tendulkar
Activate Windows
Go to PC settings to activate Windows.
You are screen sharing
Stop share
07:30 PM
17-Jun-25


TC
File Edit Run Compile Project Options Debug Break/watch
Line 5 Col 7 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char s[5][100]; int i;
clrscr();
puts("Enter 5 names ");
for(i=0;i<5;i++)gets(s[i]);
puts("NAMES");
puts("--------------------------------");
for(i=0;i<5;i++)puts(s[i]);
getch();
}
Activate Windows
Go to PC settings to activate Windows.
You are screen sharing
Stop share
07:33 PM
17-Jun-25


TC
Enter 5 names
Virat Kohli
Rohith Sharma
Ms Dhoni
Ravindra Jadeja
Shubman Gill
NAMES

Virat Kohli
Rohith Sharma
Ms Dhoni
Ravindra Jadeja
Shubman Gill
Activate Windows
Go to PC settings to activate Windows.
You are screen sharing
Stop share
07:33 PM
17-Jun-25